

Abstract No.

Time limited add-on of vinblastine to targeted therapy with selumetinib for NF1 associated pediatric low-grade glioma – a Phase 2 study proposal

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Background:

Pediatric Neurofibromatosis type 1-associated low-grade gliomas (NF1-pLGG), particularly optic pathway gliomas, represent a unique subset of pLGG due to their impact on visual outcome besides other tumor associated burden. Vinblastine first-line monotherapy has shown long-term antitumoral benefit in NF1-pLGG and has become a standard of care option. Recently, the oral MEK-inhibitor selumetinib demonstrated radiological tumor reduction as well as visual improvement in relapsed/progressive NF1-pLGG. However, the 5yr-progression-free survival was significantly lower when compared to vinblastine monotherapy. Combining MEK inhibition with vinblastine in first-line therapy may offer an advantage to preserve visual acuity, increase duration response and reduce treatment burden in this chronic disease.

Study design:

We propose a multicenter Phase 2 trial for NF1-pLGG patients. Inclusion criteria contain the NCI criteria of NF1 diagnosis and indication for antitumoral pLGG-treatment, including visual deterioration. Exclusion criteria are age below 2 years and prior antitumoral or any treatment with a MEK inhibitor. Antitumoral treatment will consist of 6 months of weekly vinblastine intravenously and oral selumetinib for 26 cycles of 28 days. The protocol will be based on the Master Protocol EU-PEARL NF1 Version 1.0.

We define following primary endpoints: radiologic response rate (RAPNO-LGG) and visual improvement rate after 6 months (Snellen/TAC2 or other equivalent measures). Secondary endpoints are mainly radiologic response rate and visual improvement rate after 12 months. Other secondary endpoints include progression-free- and overall-survival and duration of response and treatment-free interval and patient-reported outcome measures. Exploratory endpoints such as safety and tolerability and biomarker correlates complete the study design.



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